Review Questions

7/10/2026 Rishit Goyal

Q) What is an IMU?

Ans) An IMU (Inertial Measurement Unit) is a device that measures the body's motion and orientation using an accelerometer, gyroscope and a magnetometer.

- Accelerometers – measure linear acceleration (changes in speed and direction).
- Gyroscopes – measure rotational velocity (how fast something is turning).
- Magnetometers (optional) – measure the planet's magnetic field to help determine heading, like a compass.

  An IMU tells us:

- Orientation (pitch, roll, yaw)
- Angular velocity
- Linear acceleration
- Movement and position (when combined with other sensors and algorithms)

It's drawbacks are:

- Accelerometers are noisy and sensitive to vibration.
- Gyroscopes drift over time.
- Magnetometers can be disturbed by nearby metal or electronics.

Because of this, IMUs are fused with GPS, cameras, lidar, and other sensors.

Q) What do you mean by yaw and how does 9 DOF IMU reduces yaw?

Ans) Yaw is rotation around the vertical axis of an object. The magnetometer in a 9 DoF IMU measures the planets magnetic field, providing a reference. The accelerometer measures gravity direction, gyroscope measures smooth short-term rotation and magnetometer measures heading reference and together they allow the system to continually correct yaw drift. The magnetometer provides an external heading reference, reducing the drift that occurs when yaw is determined from gyroscope data alone.

Q) What do you mean by /etc/netplan?

Ans) It is a directory on Linux distributions that contains netplan configuration files used to define network settings. It is used to set up Static IP addresses, Wi-Fi connections, Ethernet interfaces and Servers.

Q) Explain what /etc/netplan does in linux distro.

Ans) It stores YAML configuration files that describe how the system's network interfaces should be configured. These files define settings like IP addresses and subnet masks, Default gateways, Wi-Fi network credentials and Network bonds. Netplan reads store network configuration files and translates them into instructions for the actual network management service that configures the machine's network interfaces.

Q) Is cloud service a virtual machine? Are all cloud services virtual machines? Differentiate between them.

Ans) A cloud service is any resource delivered over the internet by a cloud provider. This can include virtual machine, data, storage or AI services. A Virtual Machine is a software-emulated computer that runs its own operating system. A cloud service is not necessarily a virtual machine. A virtual machine is just one type of resource that cloud providers offer. All virtual machines offered through the cloud are cloud services. Not all cloud services are virtual machines.

Q) What are the advantages of docker.

Ans) Docker is a platform that packages an application together with its files into a container. This allows the application to run in the same way across platforms. Because containers share the host OS kernel, they use less memory usage and less disk space and have a faster startup. They are lighter than a virtual machine. They are very portable and can be deployed quickly. Containers can be replicated easily making them scalable. Docker's main advantage is that it packages an application and all its dependencies into a lightweight, portable, and reproducible container that runs consistently across different environments while using far fewer resources than a traditional virtual machine.

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Version Control

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What is version control

Version control is a system that records and controls the changes made in a file or project in a way that the user can access its older versions and also see when each change was made. It is an important system in software programming as changes are constantly made in the source code by a programmer.

Why is version control important

A software project has many files in which many changes are made over time by different people working on the project.

- It helps us track changes:-
  - What was changed?
  - When was it changed?
  - Who made the change?
  - Why was it changed?
- It allows access to older versions of the project:- If a recent change in the programme shows an error and the previous version needs to be restored, this can be done by using version control.
- It allows multiple coders to collaborate:- It helps developers combine their work and manage changes made by different people.
- It makes debugging easier:- By checking the history, bugs can be identified and removed.
- It allows developers to experiment:- New features can be tried on the code without the fear of loosing the original code.

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State and explain different types of version control

There are 3 types of version control.

- Local version control: Here all the versions of the project are kept in one computer. It maintains all the versions of the project instead of relying on manually named copies. It does not need internet and is easier for individual use.

- Centralized version control: Here a central server stores the main repository and developers connect to this server to access the project. Since everyone can work on a common project, it is ideal for team projects. But a server crash can affect the project because it relies on only 1 server. Giving access every time for many operations can slow down and restrict a programmers work and cause delays.

- Distributed version control: Here all the coders have the repository and its history. They make changes by connecting to central server. These changes are combined with other repositories. It provides flexibility to distributed teams

Mention the important git commands and explain them

git init - Initializes repository
git clone - Copy a remote repository
git add - Stage changes
git commit - Save changes in repository
git status - Check status of files
git log - View commit history
git branch - Create or view branches
git merge - Combine branches
git pull - Integrate remote changes
git push - Upload changes to repository